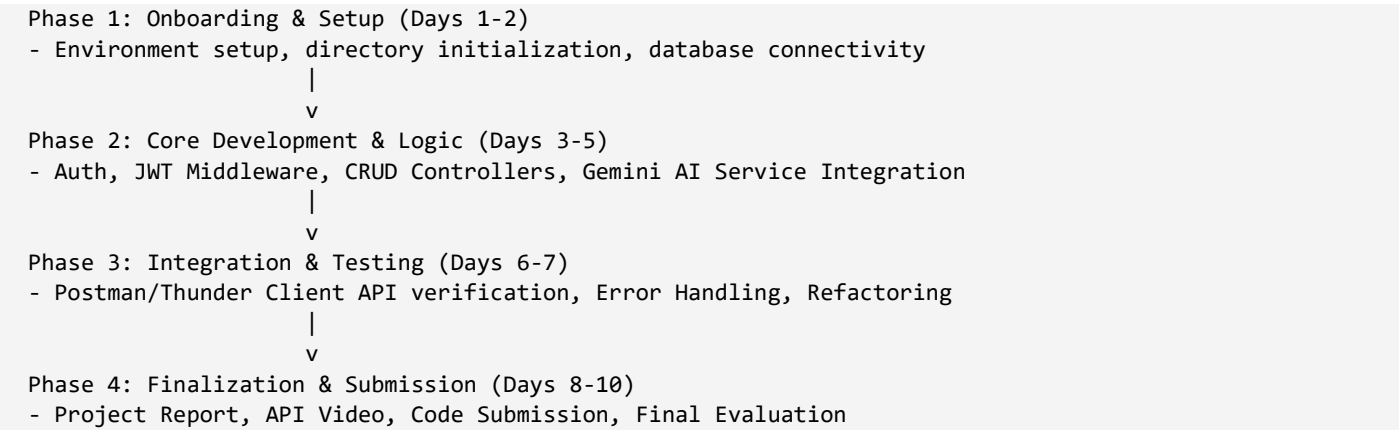


PROJECT PLANNING PHASE

1. Project Roadmap & Schedule

The development lifecycle of FitTrack AI follows a structured project timeline aligned with the academic and industry project submission milestones.



Milestone Breakdown

1. Milestone 1: Project Initialization & Onboarding

○ Initialize Node.js application, package installation, and version control.

○ Configure MongoDB Atlas connection and .env variables.
2. Milestone 2: Database Schema & Authentication Setup

○ Construct Mongoose Schemas for User and Workout collections.

○ Implement bcrypt.js password hashing and JWT authorization middleware.
3. Milestone 3: Core Business Logic & AI Service Integration

○ Develop RESTful CRUD endpoints and search filter controllers.

○ Integrate Google Gemini API service layer for workout recommendations and fitness insights.
4. Milestone 4: API Testing & Verification

○ Verify all route endpoints in Postman / Thunder Client.

○ Validate HTTP status codes, payload structures, and centralized error handling.
5. Milestone 5: Documentation & Submission

○ Record 5–10 minute project demonstration and API testing videos.

○ Push source code repository (excluding node_modules) and finalize project documentation.

2. Task Allocation (4-Member Team Distribution)

Responsibilities are distributed across team members based on backend architectural layers to ensure parallel development and smooth integration:

Team Member	Role	Primary Responsibilities	Deliverables
S. Gowtham	Team Leader & System Architect	Project setup, MongoDB connection configuration (db.js), Express server gateway (server.js), and .env security.	server.js, config/db.js, .env template, and master Git repo.
M. Gnanagiri	Auth & Middleware Developer	User registration/login business logic, bcrypt.js encryption, and JWT protection middleware.	models/User.js, controllers/authController.js, middleware/authMiddleware.js.
C. Giridharan	Workout CRUD & Search Engineer	Workout schema, full CRUD logic (Create, Read, Update, Delete), and query filtering by keyword/date/category.	models/Workout.js, controllers/workoutController.js, routes/workoutRoutes.js.

Team Member	Role	Primary Responsibilities	Deliverables
R. Hariharasudhan	Gemini AI Service & Quality Assurance	Integrating @google/generative-ai, prompt engineering for plans and insights, API testing in Postman, and video documentation.	services/geminiService.js, controllers/aiController.js, API testing video, and project documentation.

3. Environment Setup & Configuration Plan

Environment Variables Structure (.env)

All sensitive access keys, secrets, and environment parameters are isolated in a .env file at the root of the server folder:

```
# Application Server Port
PORT=5000

# MongoDB Connection String (Local or MongoDB Atlas)
MONGO_URI=mongodb+srv://<username>:<password>@cluster.mongodb.net/fittrack

# JWT Secret Key
JWT_SECRET=<your_jwt_secret>

# Google Gemini API Key
GEMINI_API_KEY=<your_gemini_api_key>

# Google Gemini Version
GEMINI_MODEL=gemini-3.7-flash
```